

Ansh Duggal

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Portfolio: <https://anshduggal.com>

TECHNICAL SKILLS

- CAD & Design: SolidWorks (3D Modelling, Assemblies, Engineering Drawings), Design for Manufacturing (DFM)
- Manufacturing & Production: Plastic Extrusion Processes, Additive Manufacturing, 3D Printing, Process Improvement, Quality Control, Production Planning
- Embedded Systems: STM32, Arduino, Raspberry Pi, Embedded Programming, Hardware–Software Integration, Debugging
- Programming: C++, Python, MATLAB
- Electronics & Control Systems: Digital Electronics, Microcontrollers, Sensors & Actuators, Control Systems, PWM, ADC, Serial Communication Protocols
- FPGA & Digital Design: Verilog, ModelSim, Quartus, Sequential Logic Design
- Engineering Tools: Microsoft Excel, Word, PowerPoint, Mbed Studio, Keil Studio

EDUCATION

University of Leeds, United Kingdom

BEng (Hons) Mechatronics and Robotics

Leeds, UK

(Sep 2022 – Jul 2025)

- Final-Year Project: Upper Limb Rehabilitation Exoskeleton for Elbow and Wrist Flexion (low-cost design with 20% weight reduction via strategic material selection).
- Relevant Modules: Electric Machines, Control Systems, Embedded Systems, Robotics & AI, Engineering Design, Additive Manufacturing.

K.R. Mangalam World School, India

Senior Secondary Education (CBSE – A-Level Equivalent)

New Delhi, India

(Mar 2019 – Jun 2022)

- Core Subjects: Physics, Chemistry, Mathematics.

ACADEMIC PROJECTS

Intelligent Upper Limb Exoskeleton for Rehabilitation

(Oct 2024 – May 2025)

- Developed a robotic exoskeleton for stroke recovery, focusing on the elbow and wrist.
- Designed a SolidWorks mechanical model and software interface to automate movement assessment and enable remote therapy.

Remote-Controlled A350 Aircraft Build (Personal Project)

(Mar 2020 – Feb 2021)

- Designed and constructed a 7ft wingspan Airbus A350 RC aircraft using Depron foam, carbon fiber, EDFs, servos, and landing gear; integrated control surfaces with an ATmega328P (Arduino Uno) microcontroller.
- Demonstrated advanced skills in prototyping, control systems integration and applied aerodynamics, reducing overall weight by 20% through strategic material selection.

Snake Game on ST Nucleo-L476RG Development Board

(Mar 2024 – May 2024)

- Created an interactive Snake game using C++ on the ST Nucleo-L476RG board, integrating peripherals like joystick, LCD, and buzzer.

FPGA-Based Dice Game

(Mar 2024 – May 2024)

- Formulated a dice game using Verilog and sequential logic, simulating and testing functionality with ModelSim and modular test benches.

Microcontroller-Controlled Washing Machine Prototype

(Mar 2023 – May 2023)

- Built a washing machine control panel using the STM32L476RG microcontroller with features like power control, cycle selection, and serial monitoring.

WORK EXPERIENCE

Brite Industries,
Operations Coordinator

New Delhi, India
(Aug 2025 –Present)

- Gained hands-on experience in plastic extrusion manufacturing, supporting the production of flexible conduit pipes from HDPE raw materials.
- Worked closely with production personnel to streamline manufacturing workflows and support continuous improvement initiatives.
- Assisted with production planning, material handling, and inventory monitoring to ensure efficient utilization of raw materials.
- Prepared production reports and operational documentation to support process tracking and performance analysis.

VOLUNTARY EXPERIENCE

Leeds RAG,
Secretary

Leeds, UK
(Jun 2023 – Jun 2024)

- Negotiated contracts with catering and equipment vendors for fundraising events, securing 8% cost savings, and boosted sponsorship revenue by 22% through strategic outreach.

Interact Club, Rotary International,
Member

New Delhi, India
(Apr 2018 – Apr 2019)

- Coordinated blood donation drives, benefiting 500+ individuals, and partnered with local organizations for community outreach.

LEADERSHIP ACHIEVEMENTS

Head Boy,
K.R. Mangalam World School

New Delhi, India

- Directed a team of 20 prefects, increasing event participation by 75% and enhancing engagement.
- Spearheaded initiative to create a student-faculty mentorship program, matching 50+ students with faculty mentors resulting in a 20% increase in student engagement in research opportunities.